

Catalytic Contact Time Lock for Amyloid Beta Clearance

Executive Technology Brief

Domain: Enzymatic Catalysis + Mass Transport

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Amyloid beta accumulation in CSF of Alzheimer's/NPH patients with shunts

Mechanism: NEP enzyme immobilized on catheter wall cleaves A-beta-42 via mass-transport-limited contact time

Architecture: Catheter with immobilized neprilysin (NEP) enzyme on inner lumen wall; CSF flows past, A β 42 in CSF contacts enzyme for sufficient time (contact-time lock) to undergo catalytic cleavage

Governing Model: Mass-transport-limited enzyme catalysis in catheter lumen. Reaction rate is bounded by either enzyme kinetics (Michaelis-Menten) or convective delivery of substrate to enzyme surface, whichever is smaller.

Key Equations:

$v = (V_{max} * [S]) / (K_m + [S])$ [enzyme kinetics]

$J = -D * dC/dx$ [diffusive flux]

$Da = (k * L) / D$ [reaction vs transport]

$Pe = (v * L) / D$ [Péclet number, convective vs diffusive transport]

$k_{cat_obs} = k_{cat} * (1 - deactivation(t))$ [enzyme deactivation over time]

Kill Condition: Enzyme half-life in CSF environment is less than 30 days (insufficient for chronic implant)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.